TAHIR SAJJAD BUTT

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EDUCATION

Lahore University of Management Sciences (LUMS), School of Science and Engineering (SSE), Lahore, Pakistan
MS Physics Sep 2022 – June 2024

CGPA: 3.56/4.00
Relevant Coursework: Quantum Mechanics, Electrodynamics, Condensed Matter Physics, Advanced Statistical Mechanics, Photovoltaic Devices, Computational Physics, and Quantum Information Science & Technology.

University of Management and Technology (UMT), School of Sciences, Lahore, Pakistan
BS (Hons) in Physics Oct 2018 – July 2022
CGPA: 3.29/4.00
Relevant Coursework: Classical Mechanics, Quantum Mechanics, Electromagnetic Theory, Thermal and Statistical Physics, Nuclear Physics, Digital Electronics, Solid State Physics, Laser Physics and Nanotechnology.

JOURNAL PUBLICATIONS

1. Iqbal, S., Akbar, F., Salar, M. A., **Butt, T. S.**, Khan, A. A., Sohail, M., & Yameen, B. (2025). Demonstrating donor–acceptor covalent conjugate of polythiophene and perylene diimide as an all-organic photocatalyst for enhanced light-driven H_2O_2 production. ACS Applied Materials & Interfaces**,** 17(19), 28609–28621. https://doi.org/10.1021/acsami.4c22745
   Contribution: fabricated the P3HT, P3HT-PDI-PH, and PTh-PDI-DCP photodetectors; built the chopped-light optical setup; measured and analyzed transient photocurrent, including rise and fall times (Fig. 3).
2. Jawad, M., Siddique, S., Aslam, Z., Hussain, S., **Butt, T. S.**, Saeed, F., Ain, Q. U., Ulhaq, A., & Khan, A. A. (2025). Temperature-dependent photoluminescence and grain size control in a prototypical lead-free bismuth halide perovskite. Advanced Engineering Materials, 27(5), 2402009. https://doi.org/10.1002/adem.202402009
   Contribution: fabricated $MA_3Bi_2I_9$ thin films with anti-solvent grain-size control (Fig. 2, Table 1); fabricated photodetectors and measured their photocurrent response (Fig. S2, Supporting Information).

RESEARCH EXPERIENCE

Functional Materials and Optoelectronic Devices (FMOD) Lab, LUMS, Lahore, Pakistan
Research Assistant May 2024 – Aug 2026
Supervisor: Dr. Ammar Ahmed Khan
Project: Fabrication and Time-Dependent Analysis of Polymer Photodetectors

- **Fabrication:** Fabricated mesoscopic photodetectors (FTO/compact TiO_2/mesoporous TiO_2/polymer/Cr-Au) by spin coating, using P3HT, the P3HT–PDI physical hybrid (P3HT-PDI-PH), or the covalent donor–acceptor conjugate PTh-PDI-DCP as the active layer; deposited Cr-Au contacts by thermal evaporation.
- **Characterization:** Measured transient photocurrent under 0.33 Hz chopped light at 0 V bias with a transimpedance amplifier; extracted 10–90% rise and fall times by curve fitting.
- **Results:** PTh-PDI-DCP gave the shortest rise time (109 ± 3 ms), compared with P3HT-PDI-PH (133 ± 2 ms) and P3HT (172 ± 37 ms), while fall times did not show this speed-up; the fall times of PTh-PDI-DCP and P3HT-PDI-PH were longer than their rise times. The paper hypothesizes that the covalent bridge speeds charge transport from donor to acceptor, giving faster injection into the TiO_2 scaffold. Published in [1] (Fig. 3).

Functional Materials and Optoelectronic Devices (FMOD) Lab, LUMS, Lahore, Pakistan

Graduate Research (MS Thesis) June 2023 – May 2024
Supervisor: Dr. Ammar Ahmed Khan
Title: A Photodetector Based on Lead-Free Perovskite-like Methylammonium Bismuth Iodide (MBI)

- **Fabrication:** With a team, grew $MA_3Bi_2I_9$ (MBI) thin films by one-step spin coating from a methylammonium iodide/bismuth iodide precursor, varying the chlorobenzene anti-solvent from 20 to 175 $\mu L \cdot cm^{-2}$ to control nucleation. Independently fabricated mesoscopic photodetectors (FTO/compact TiO_2/mesoporous TiO_2/MBI/Cr-Au) at 90, 60, and 45 $\mu L \cdot cm^{-2}$.
- **Characterization:** Analyzed SEM and XRD data to confirm the hexagonal MBI phase (space group $P6_3/mmc$) and quantify grain-size trends; designed the photodetector test assembly and measured device response under chopped white light (0 V bias, transimpedance-amplifier readout).
- **Results:** The amount of anti-solvent tuned the average grain size by about 26-fold (1.40 ± 0.03 to 36 ± 1 µm). Device performance peaked at the intermediate anti-solvent concentration: reducing the anti-solvent from 90 to 60 $\mu L \cdot cm^{-2}$ raised peak photocurrent from about 2.6 to 7 $\mu A \cdot cm^{-2}$ and gave the fastest rise time (0.35 s), while 45 $\mu L \cdot cm^{-2}$ lowered photocurrent, attributed to larger but more sparsely spread grains. Published in [2] (Fig. 2, Table 1, and Fig. S2).

Physical Vapor Deposition Lab, Center for Advanced Studies in Physics (CASP), Government College University (GCU), Lahore, Pakistan
Bachelor's Thesis Research (external thesis placement) July 2021 – June 2022
Supervisor: Hira Yaseen
Title: Influence of Repetition Rate on the Growth of MgO Thin Film by Pulsed Laser Deposition Technique

- **Fabrication:** Operated a Q-switched Nd:YAG pulsed laser deposition system (1064 nm, 200 mJ pulse energy, 10 ns pulse width) to grow MgO thin films on silicon substrates under vacuum (10^{-3} Torr), varying the laser repetition rate (2, 3, 4, and 5 Hz), with 2000 to 5000 total laser shots.
- **Characterization:** Analyzed XRD, SEM, and AFM data to determine film thickness, crystallite size (Scherrer method), and grain morphology at each repetition rate.
- **Results:** Film thickness rose from 307 nm (2000 shots) to 960 nm (5000 shots), while crystallite size fell from 65 to 15 nm.

ACADEMIC PROJECTS

Device Simulation of a Lead-Free MBI Solar Cell (TCAD Silvaco) Jan 2024 – April 2024
Coursework Project

- **Design:** Built an n-i-p $MA_3Bi_2I_9$ (MBI) solar-cell model in TCAD Silvaco, defining the layer stack, material parameters, mesh, and boundary conditions.
- **Simulation:** Simulated photogeneration and recombination-rate profiles, quantum efficiency, and J–V characteristics under different illumination conditions; extracted V_{OC}, J_{SC}, V_{mp}, J_{mp}, fill factor (FF), and power conversion efficiency (PCE).

Quantum Key Distribution: BB84 and B92 Protocols (GitHub) Jan 2023 – May 2023
Coursework Project

- **Implementation:** Implemented the BB84 and B92 quantum key distribution protocols in Python with Qiskit on the Aer simulator, including an eavesdropper (Eve) model showing that interception is detectable.
- **Analysis:** Wrote a report analyzing both protocols, with histogram visualizations of the simulated key exchange.

TEACHING EXPERIENCE

School of Advanced and Global Education (SAGE), Lahore, Pakistan Aug 2026 – Present

STEM Lab Developer & Instructor

- **Curriculum design:** Designing an interdisciplinary STEM curriculum for Grades 6–8 built around three modules (Pipes & Systems, Power & Purpose, and Wheels & Mobility) that turn core physics and engineering concepts into hands-on, age-appropriate activities.
- **Laboratory setup:** Setting up the program's teaching laboratory from scratch, including equipment selection, procurement, and experimental stations, so that students learn engineering concepts through direct experimentation.
- **Program development:** Introducing project and experiment-based learning through original hands-on activities in which students design, build, test, and troubleshoot.

IB Prep Academy, Lahore, Pakistan Aug 2025 – Present

MYP & DP Physics Instructor (Part Time)

- **Physics teaching:** Teaching IB Physics to MYP 4–5 and DP 1–2 students following the IB curriculum at IB Prep Academy.
- **Teaching methodology:** Focusing instruction on conceptual clarity and exam technique, tailoring approach to each student's problem-solving gaps rather than rote content delivery.

TECHNICAL SKILLS

Fabrication	Spin coating, anti-solvent crystallization, pulsed laser deposition (PLD), thermal evaporation of metal contacts, plasma cleaning
Characterization	I–V measurements, transient photocurrent (chopped light, transimpedance amplifier), photodetector test-assembly design, digital storage oscilloscope, source meter, PhysLogger (data logger); XRD, SEM, AFM, and UV-Vis data analysis
Simulation	TCAD Silvaco, SCAPS-1D
Programming	Python (NumPy, Pandas, Matplotlib, Seaborn), Qiskit, MATLAB, Mathematica
Tools	LaTeX, MS Office, Origin (OriginLab).

AWARDS AND HONORS

Dean's Merit Award, UMT Oct 2021 – June 2022

- Awarded in two semesters of the 2021–22 academic year for a semester GPA of 3.80/4.00 or higher.

LEADERSHIP AND OUTREACH

Vice President, SPIE Student Chapter, LUMS June 2023 – Dec 2024

- Organized interactive optics and light-physics demonstrations at local schools, bringing hands-on science outreach to young and underprivileged students.
- Helped run chapter activities in optics and quantum science, including initiatives that built student engagement with computational and quantum optics at LUMS.

Vice President, Ibn-Al-Haytham Physics Society (IPS), UMT July 2021 – Aug 2022

- Organized seminars and workshops for physics students at UMT.